

enterprise resource planning (ERP) systems. However, these are used for non-core activities such as inventory, ordering and invoicing. Core manufacturing processes are still labour-intensive. Use of CNC machines, programmable logical control (PLC) or supervisory control and data acquisition (SCADA) systems for automation are not common. Data from these systems remain in the common pool and are rarely used for process improvements.

The government of India's Make in India initiative is designed to help manufacturing industries grow and compete in global markets. One such effort by Ministry of Heavy Industries is called *Samarth Udyog Bharath 4.0*. It is aimed to make Indian industries ready for Industry 4.0. It helps bring together industries in India with IIoT solution providers.

The IIoT is transforming manufacturing conditions in India. Apple India recently established a manufacturing IoT system, which is driving Industry 4.0. It aims to accelerate the industry to a new level, to transform economies and to open a new era of growth and competitiveness.

Tech Mahindra has been a member of General Electric (GE) Digital Alliance programme since 2016. The initiative is working towards building a digital ecosystem.

Tech Mahindra is also an early adopter of Predix platform, which is GE's cloud-based platform as a service (PaaS) for the industrial Internet. Predix connects Tech Mahindra's assets with data and analytics, with unprecedented speed and scale. It creates innovative solutions to address the needs of power, oil and gas, and transportation industries, among others.

In addition, Tech Mahindra is working on building solutions for smart city buildings. With over 100 Predix-trained experts, the company plans to invest in more than 1000 developers and architects for Predix platform in the next few years.

Connecting the IIoT to the automotive sector

As manufacturing and automotive sectors across the globe are getting ready



Smart meter is an important application of the IIoT (Credit: <https://enterpriseiotinsights.com>)

for Industry 4.0, India is also gearing up for this change. This shift in the automotive and transportation industry benefits India immensely and will bring an entirely different paradigm.

In 2006, German government came up with Industry 4.0 to digitalise the manufacturing sector to increase productivity. While Industry 3.0 was simply an automation of isolated machines, Industry 4.0 demonstrated the timely use of all physical assets of the factory and their integration into the digital ecosystem with value chain partners.

Industry 4.0 is motivating technologists and entrepreneurs to develop new solutions with technological spins. A strong cyber security infrastructure is slowly building up in India, which is a major requirement for the IIoT's capabilities.

By 2020, India is expected to become a major automobile manufacturing hub and the third largest market for automobiles in the world. As such, the automobile industry in India needs a dramatic increase in their level of digitalisation and investment in data analytics. As a result, IIoT technology has become a key enabler for bringing out change, which is driving a positive impact beyond the industry.

Industry 4.0 has its own road to follow for smart manufacturing. It needs support from segments like 3D printing, 3D sensors, social software, augmented reality (AR) and location awareness to enter the next era of smart production.

Automakers in India have already made headway into Industry 4.0.

Bosch has initiated the implementation of Industry 4.0 in all of its 15 manufacturing facilities across India. The company is adopting national initiatives such as Make in India and Digital India to help its business move faster to market. It has collaborated with GE Digital for connected manufacturing and with SAP to develop cloud technology and software solutions.

Using the IIoT as a medium in the power sector

The IIoT provides great opportunities in the power sector across Asia-Pacific. GE predicts that US\$ 1.3 trillion can be captured in the electricity value chain from 2016 to 2025 globally by the IIoT. Smart meters and smart thermostats are two important recognisable consumer-oriented applications of the IIoT in the power sector.

Older smart meters were Internet-connected devices for sending electricity consumption data to the concerned power utility. In newer versions, a variety of add-on services have been added. These services include energy management portal, net-metering, pre-paid electricity purchase, inventory control identification, voltage management to reduce losses and so on.

Burbank in California has benefitted greatly from the installation of advanced metering infrastructure. It has enabled about two per cent reduction in electricity usage per customer. Further, it has reduced response time to customer inquiries, brought 87 per cent reduction in field visits, improved reliability of grid with drop in system failure and more.

Smart thermostats are Internet-connected devices for measuring temperature and humidity inside homes or offices. These send collected data to the cloud. Smartphones can now do the job using the IIoT.

In addition, many machine learning applications have been developed for saving energy and user comfort. Applications of the IIoT in power sector mainly focus on power-generation plants (atomic, thermal, solar, wind, hydro-electric, etc) and in transmission and distribution grid.

In Asia, many grids are often unreliable. This is due to aged equipment,